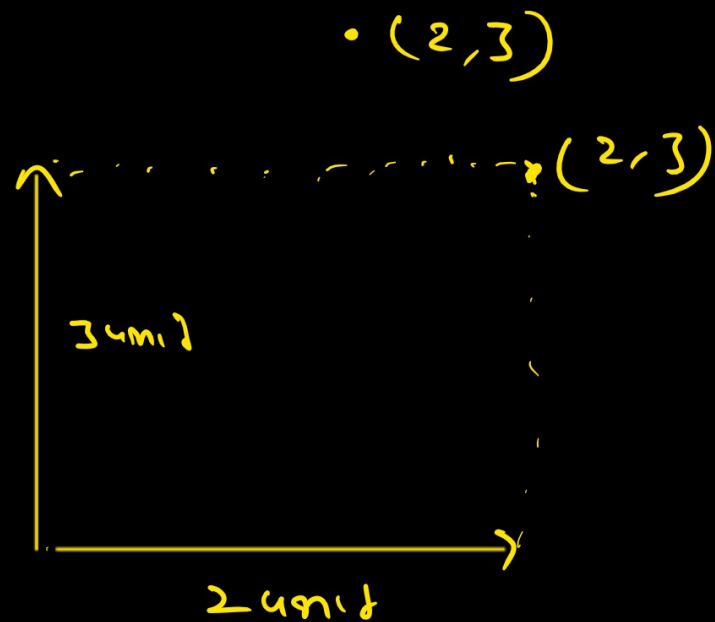


Vector :-

$$\vec{v} = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \rightarrow \text{Study Hour} \\ \rightarrow \text{Sleep Hour}$$



Row Vector = $[2, 3, 4]$

$$\text{Col Vec} = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$$

$$\vec{a} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \vec{b} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

$$\vec{a} + \vec{b} = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$$

$$\vec{a} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad b = 5$$

$$\vec{a} \times b = \begin{bmatrix} 5 \\ 10 \end{bmatrix}$$

$$A = \begin{matrix} \rightarrow \\ \rightarrow \end{matrix} \begin{bmatrix} \overset{\downarrow \text{C}}{\underset{\downarrow \text{D}}{1}} & 2 \\ 3 & \underset{\downarrow \text{D}}{4} \end{bmatrix}_{2 \times 2}$$

$$\begin{bmatrix} \cancel{1} & 2 \\ \cancel{3} & \cancel{4} \end{bmatrix}_{2 \times 2}$$

$$= \begin{bmatrix} \underline{1 \times 1 + 2 \times 1} & 1 \times 2 + 2 \times 1 \\ 3 \times 1 + 4 \times 1 & 3 \times 2 + 4 \times 1 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 4 \\ 7 & 10 \end{bmatrix}$$

$$A_{3 \times 2} \times B_{2 \times 2} = \begin{bmatrix} 1 & 2 \\ 1 & 1 \\ 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 \times 1 & 2 \times 1 \\ 1 \times 1 & 2 \times 1 \\ 1 \times 1 & 2 \times 1 \end{bmatrix}$$

$$A_{3 \times 2} \times B_{2 \times 5} = C_{3 \times 5}$$

$$A_{3 \times 2} = \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 5 & 1 \end{bmatrix}$$

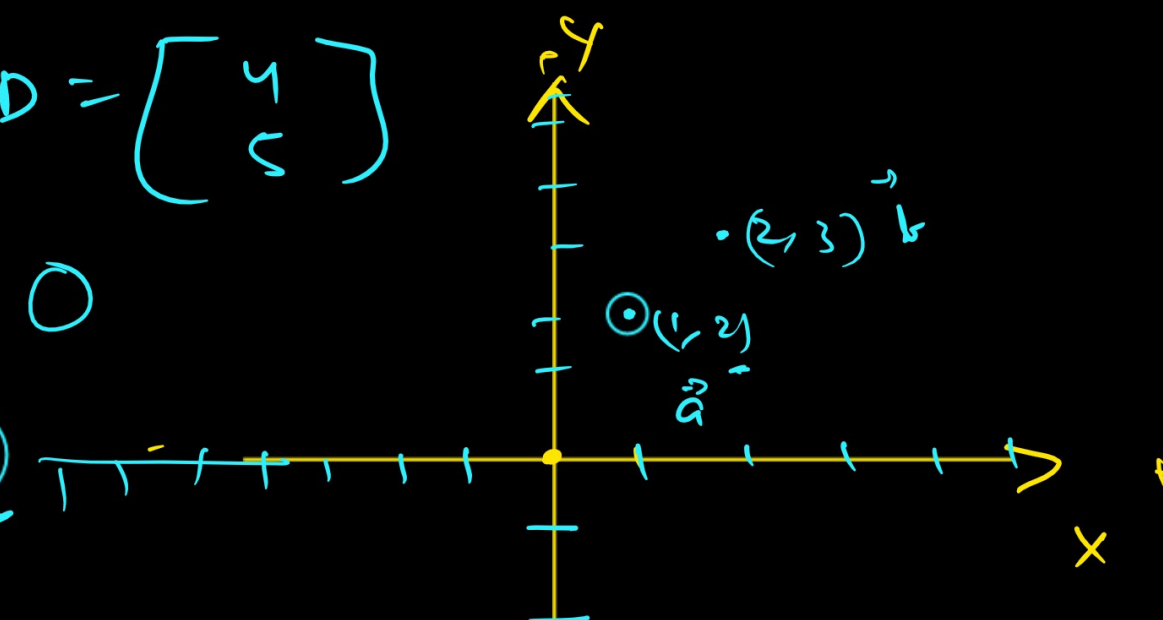
$$B_{2 \times 5} = \begin{bmatrix} 1 & 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} - & - & - & - & - \\ - & - & - & - & - \\ - & - & - & - & - \end{bmatrix}$$

① $\vec{A} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, ~~$\vec{B} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$~~

$D = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$

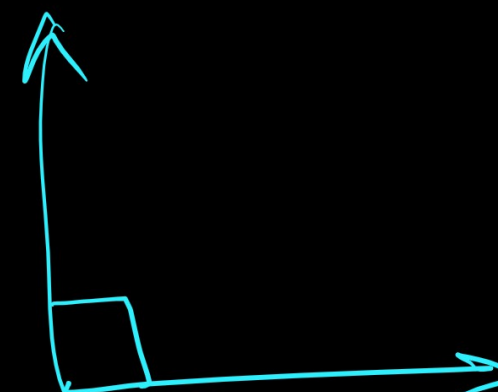
$\vec{A} \cdot \vec{D} = 1 \times 2 + 2 \times 3 = \textcircled{8}$



② $\vec{a} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ $\vec{c} = \begin{bmatrix} -10 \\ -2 \end{bmatrix}$

$= -10 + (-4)$
 $= -14$

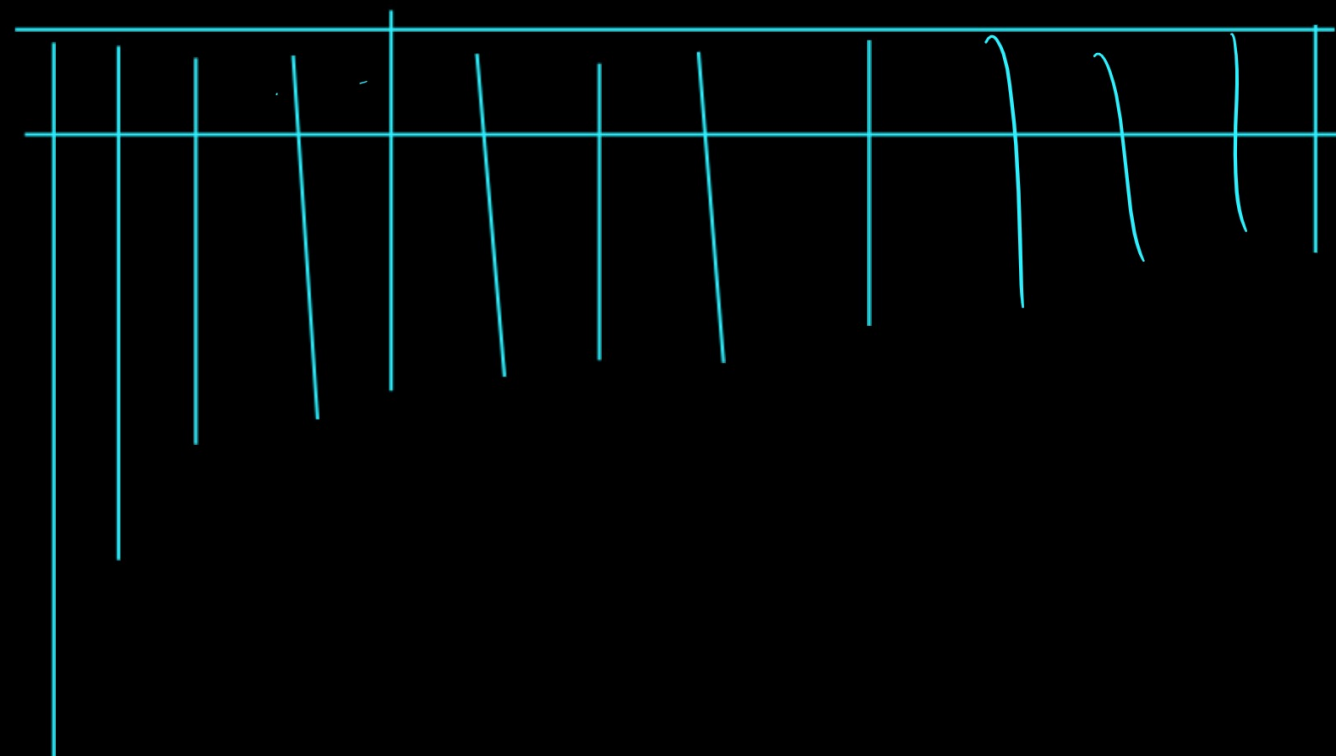
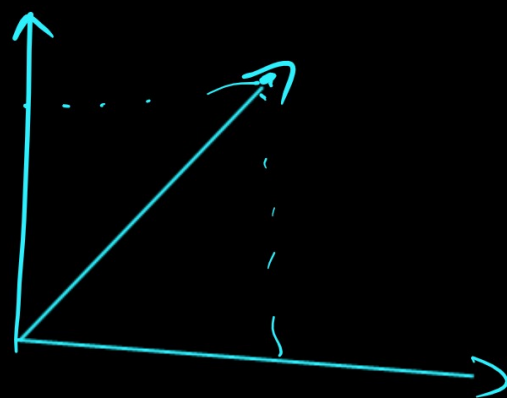
$\textcircled{c}$
 $(-10, -2)$



$$\cos(\theta) = \left(\frac{\vec{a} \cdot \vec{r}}{|\vec{a}| |\vec{r}|} \right)$$

$$\vec{a} \cdot \vec{r} = (19 \mid 16 \mid \cos \theta)$$

$$\begin{bmatrix} 5 \\ 6 \end{bmatrix}$$



$$\vec{v} = \lambda \vec{v}'$$

$$1000 \times 1000$$

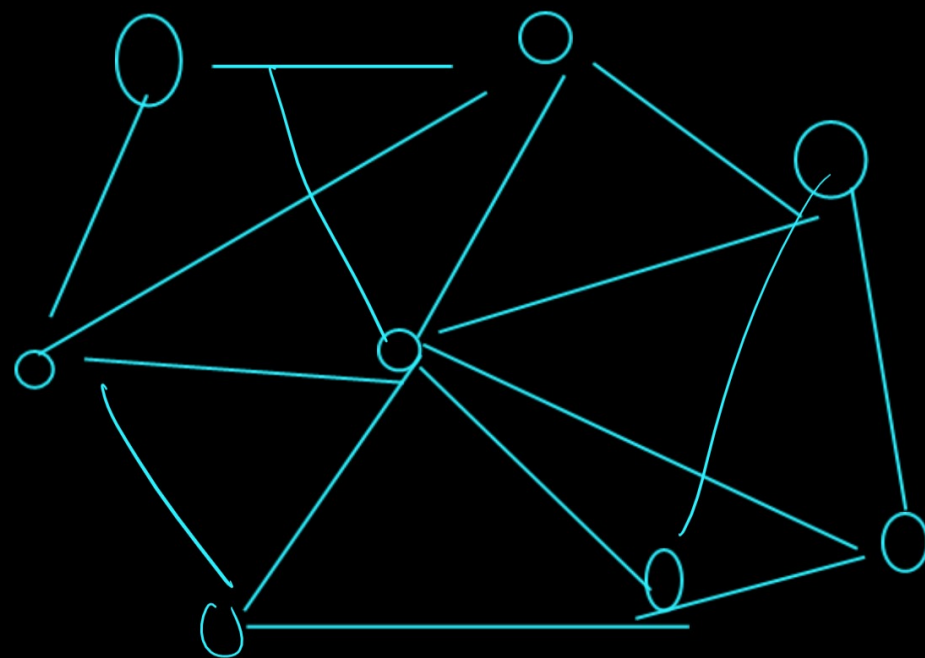
$$A \vec{v} = \lambda \vec{v}$$

$$(2000 \times 2000)$$

$$A = \left[\begin{array}{c|c} \begin{array}{c} 2 \\ 0 \end{array} & \begin{array}{c} 1 \\ 3 \end{array} \end{array} \right]$$

$$\begin{aligned} \textcircled{2} \times \begin{bmatrix} 1 \\ 0 \end{bmatrix} &= \begin{bmatrix} 2 \\ 0 \end{bmatrix} \\ \textcircled{3} \times \begin{bmatrix} 0 \\ 1 \end{bmatrix} &= \begin{bmatrix} 0 \\ 3 \end{bmatrix} \end{aligned}$$

$$\text{2D} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$



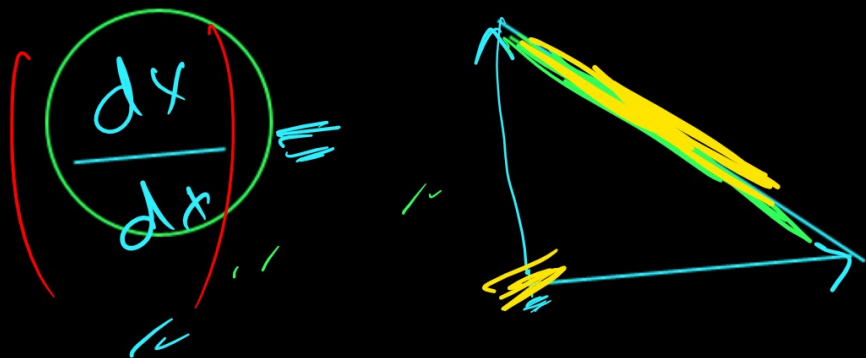
$$y = x \quad \checkmark$$

$$y = f(x) = x^2 + 1$$

$$y = 2x$$

$$= x + x^2 + 1$$

$$y = f(x)$$



$$y = x^2$$

$$\frac{dy}{dx} = 2x$$

$$y = 2x$$

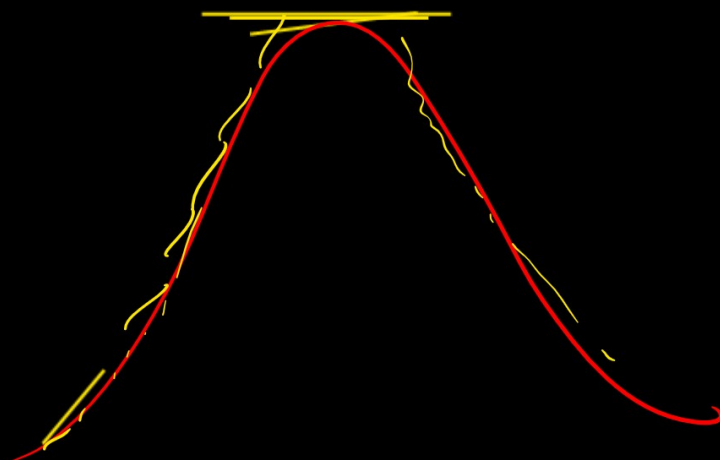
$$x = 1 \quad \text{slope} = 2$$

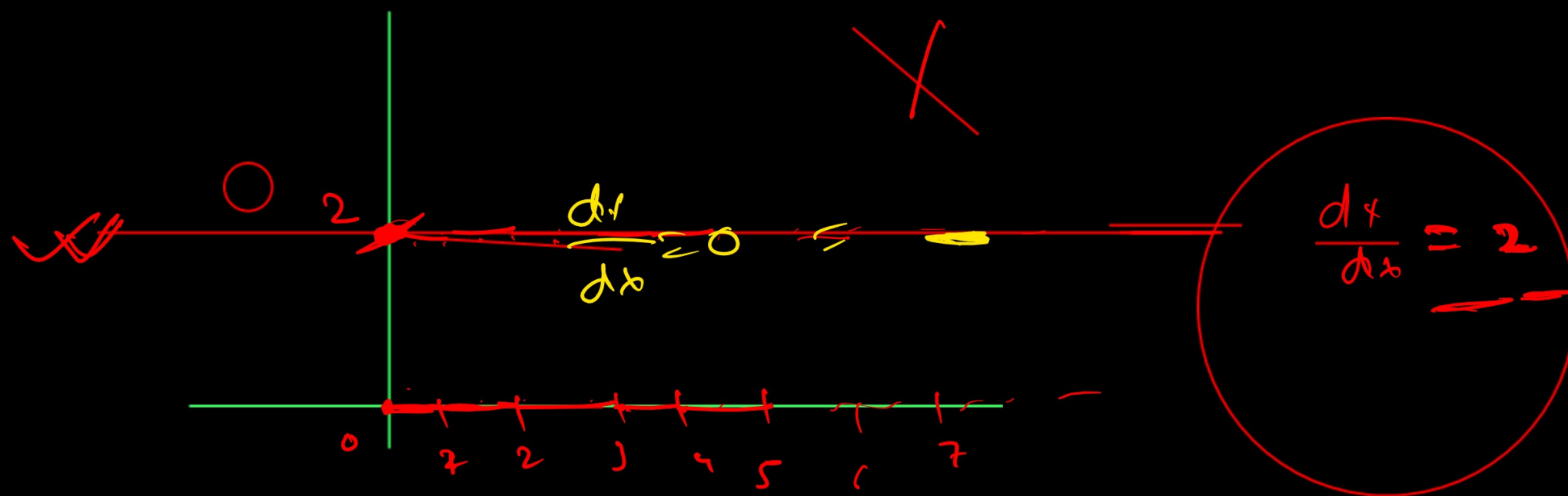
$$x = 3 \quad \text{slope} = 6$$

$$\left(\frac{dy}{dx} \right) = 2$$

$$y = 2x$$

$$\left(\frac{dy}{dx} \right) = 2 \quad \checkmark$$



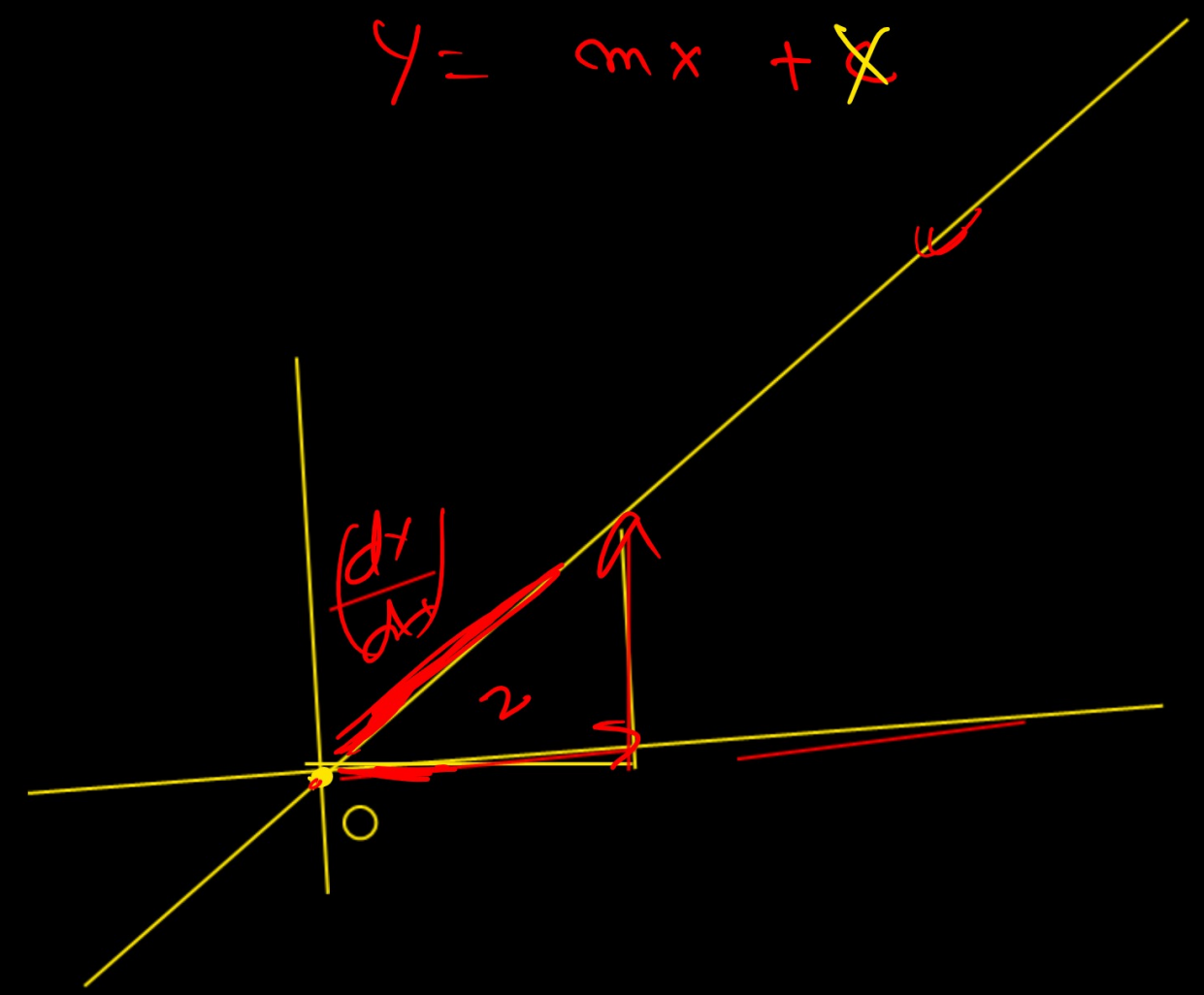


$$y = mx$$

$$y = 2x$$

$$y = mx + x$$

$$(y = 2x + 2)$$



$$y = 2x + 2$$

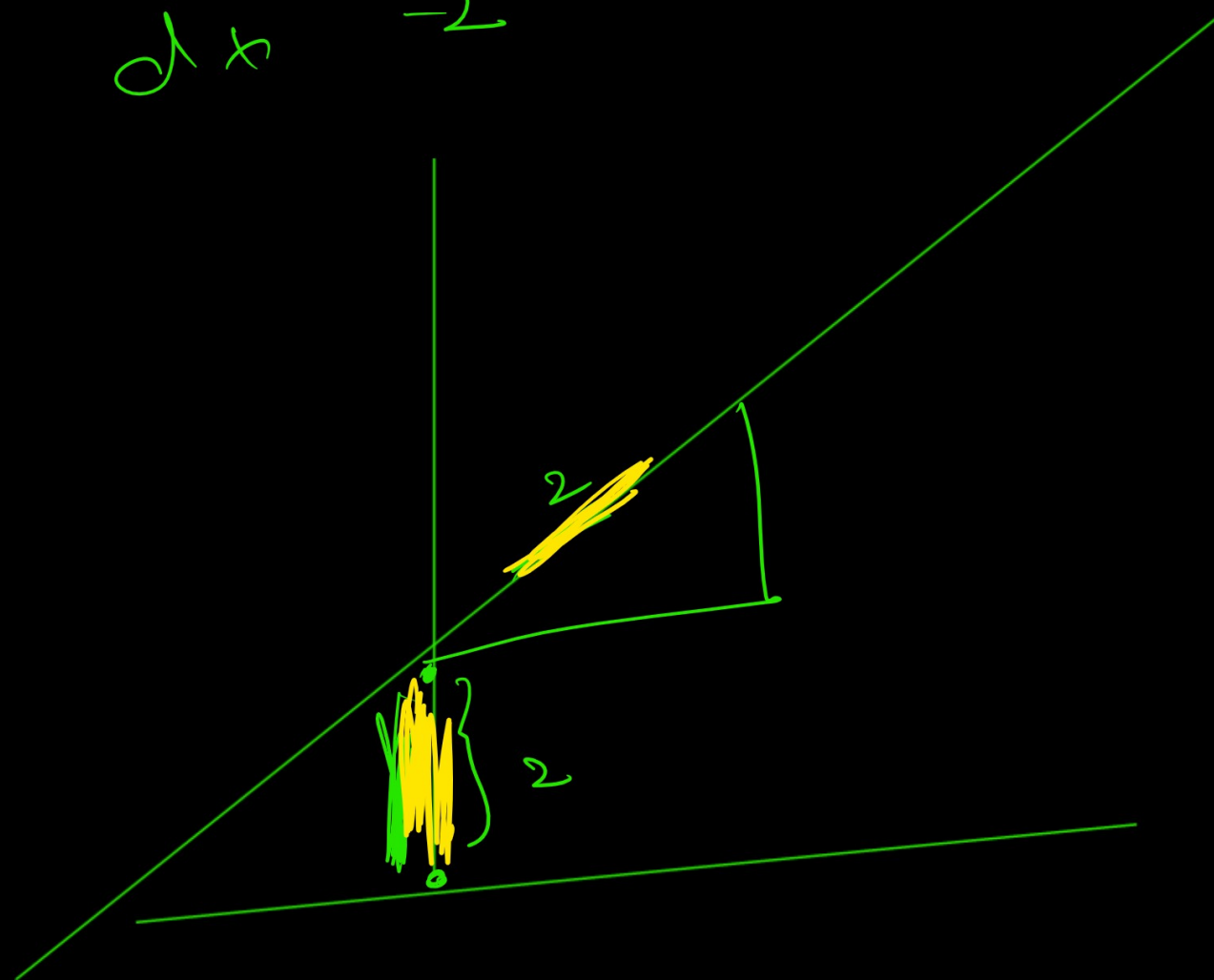
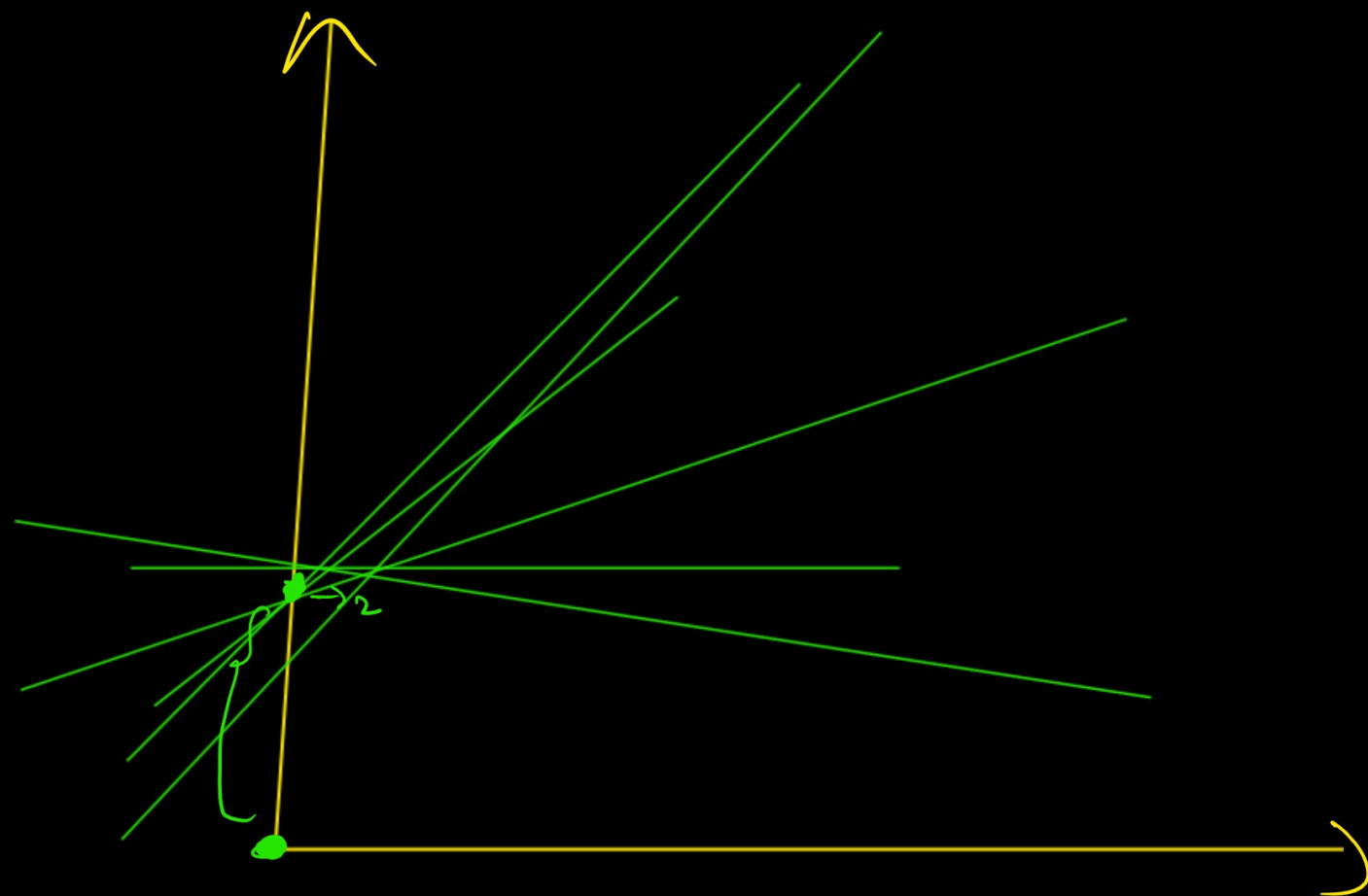
$$y = 2$$

$$x = 0$$

$$y = mx + c$$

$$\frac{dy}{dx} = 2$$

$$= 2$$



$$\textcircled{1} \quad \underline{y} = \underline{x}^2 \quad \frac{dy}{dx} = 2x$$

$$c = \text{const}$$

$$\underline{z} = f(\underline{x}, \underline{y}) = \underline{x}^2 + \underline{y}^2$$

$$\frac{\partial z}{\partial x \partial y} \quad \frac{\partial z}{\partial x} \quad , \quad \frac{\partial z}{\partial y}$$

$$\frac{\partial z}{\partial x} = 2x$$

$$\frac{\partial z}{\partial y} = 2y$$

$$\nabla z = \begin{bmatrix} 2x \\ 2y \end{bmatrix}$$

Chain Rule

$$y = f(g(x))$$

$$\frac{dy}{dx} = \left(\frac{dy}{dg} \cdot \frac{dg}{dx} \right)$$

$$\underline{g(x)} = \underline{2x} \quad \frac{\underline{dg}}{\underline{dx}} = 2$$

$$y = x^2$$

$$\underline{y} = \cancel{f(\underline{g})} = g^2$$

$$\frac{dy}{dg} = 2g$$

$$\boxed{\frac{dy}{dx} = \left(\frac{dy}{dg} \right) \cdot \left(\frac{dg}{dx} \right)}$$

$$\left(\frac{dy}{dx} \right) = 8x$$

$$= 2g \cdot 2$$

$$= 4g$$

$$= 4 \times 2x$$

$$= 8x$$